

MAZEX'24

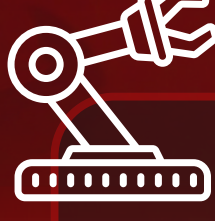
ROBO DE DINERO

Greetings, future masterminds! Welcome to your first event under the Robotics Club, "MazeX." This year's thrilling theme is "ROBO DE DINERO". Get ready to guide your autonomous robot, codenamed "The Professor," on an epic mission to infiltrate a high-security bank, avoid guards, and successfully retrieve the treasure. Along the way, you'll navigate through tricky obstacles, make strategic decisions, and prove your heist skills in this exciting challenge.



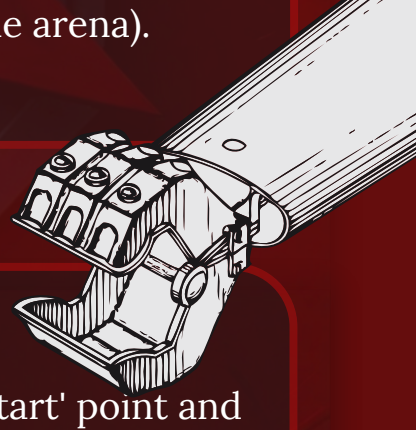
MISSION BRIEF

In this year's ROBO DE DINERO mission, your autonomous robot, "The Professor," has a dual objective. Your robot must first navigate through the high-security bank, evading guards. The ultimate goal is to break into the vault and retrieve the treasure in the minimum time, while also planning the safest escape route to avoid capture. Timing and precision are key to a successful heist!



ARENA SPECIFICATION

The heist arena is a 270cm x 270cm square-shaped environment, designed to simulate a high-security bank layout with intricate paths and hidden challenges. A black strip measuring 30mm wide represents the critical route that your line-following robot, "The Professor" must follow to navigate through the bank's corridors. Along the way, your bot must carefully evade obstacles representing security guards, each being a box measuring 60mm x 60mm x 150mm, to reach the treasure(indicated by a black rectangle in the arena).



GAMEPLAY

Your bot, "The Professor," must start from the 'Start' point and navigate its way through the bank to reach the 'End' of the arena. The bot must follow an algorithm to discover the correct path, avoiding obstacles (guards) along the way. As "The Professor" navigates the bank, it can store the turns in its memory to optimize the return journey, identifying the shortest and safest route. Once the treasure is picked up, your bot must return to the starting point, avoiding any paths where guards were previously encountered.

You have a total of 5 minutes to complete the heist—grab the treasure and return to safety. If your bot takes longer than 5 minutes, the extra time taken will be deducted from your overall score, so speed and precision are essential!



BOT SPECIFICATIONS

"The Professor" must fit within a box of dimensions 220mm x 220mm x 220mm. The robot must not cause any damage to the bank arena, leave residue, or create markings—any such violations or damage will lead to disqualification. The bot must be powered by an internal source with a voltage rating not exceeding 24V.

Additionally, the machine cannot be constructed using pre-made 'Lego Kits' or any pre-assembled mechanisms, although pre-made gear assemblies are allowed for use.

RULES AND SCORING METHODS

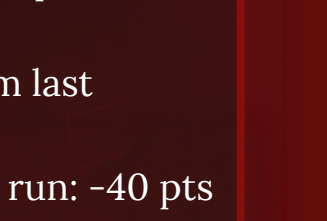
- Bot Construction: 100 pts
- Reached Vault Successfully: 100 pts
- Escaped from the bank with money successfully: 100 pts

NEGATIVE POINTS

- Caught by Police(hits obstacle):-20 pts + restart from last junction
- Enters path with Police (obstacle) during the return run: -40 pts

TIME POINTS

- (Remaining Time in seconds/300 seconds) *300 pts



TEAM SPECIFICATIONS

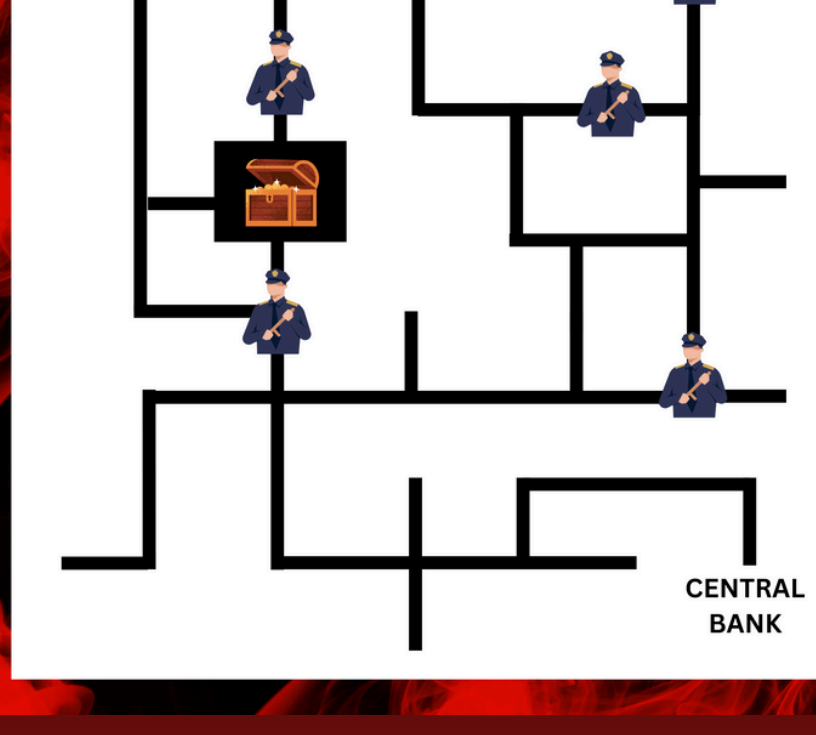
Teams can consist of up to 4 members(including team leader), and can include members from different branches.

CERTIFICATION POLICY

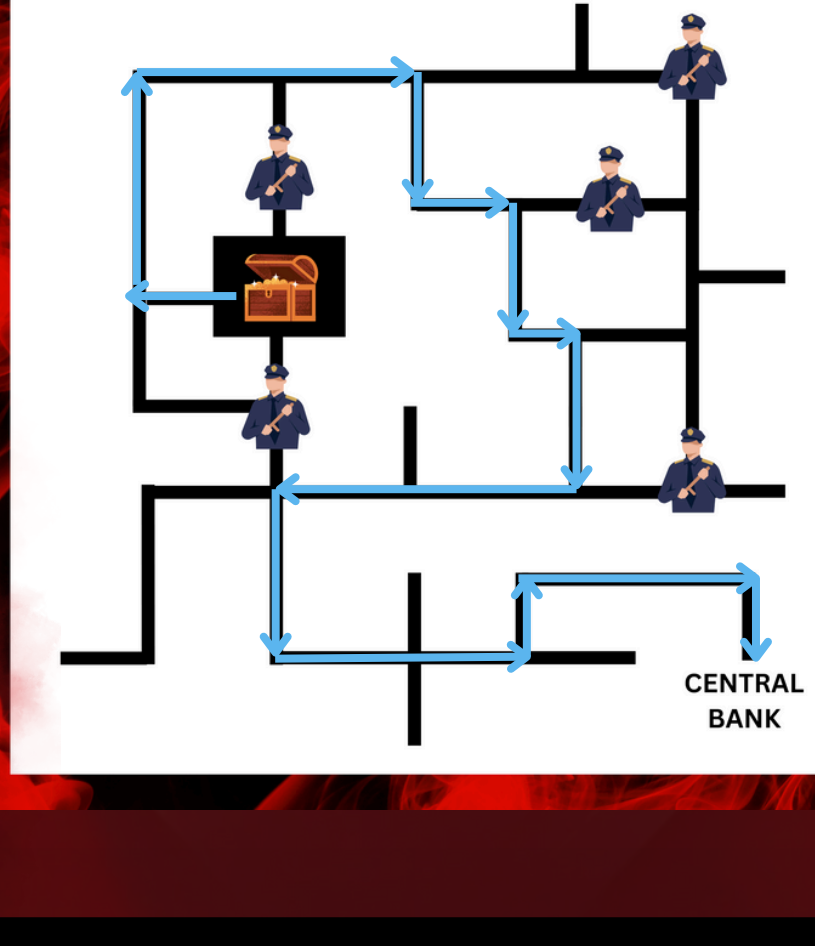
- The top three teams will receive prizes along with certificates of excellence.
- All participating teams from will receive certificates of participation.



HERE IS AN EXAMPLE LAYOUT OF THE MAZE



AFTER PICKING UP THE MONEY THE BOT WILL NEED TO RETURN THROUGH THE MOST OPTIMAL PATH AVOIDING THE POLICEMEN...



TO PARTICIPATE IN MAZEX, SCAN THE QR CODE.



ALL THE BEST !!

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